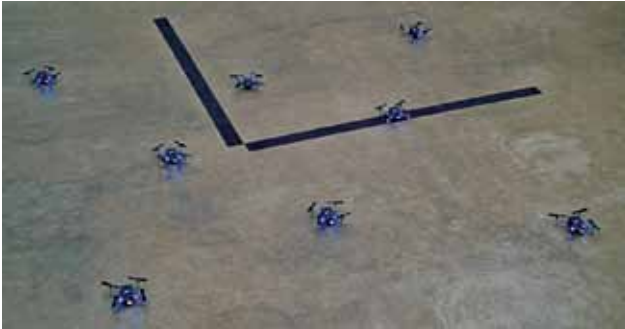




Training method developed for drones' safe operation in crowded places

MIT engineers have developed a training method for multiagent systems, ensuring safe operation in crowded environments. Using GCBF + (graph control barrier func-



MIT engineers developed a training method for multiagent systems, such as large numbers of drones, that can guarantee their safe operation in crowded environments (Credit: Courtesy of the researchers)

tion), agents learn to map dynamic safety margins, avoiding collisions while adapting to larger groups. In tests with palm-sized drones, the system enabled midflight position changes and landing on moving vehicles. The method scales effectively, requiring only a few agents to train, making large-scale, coordinated operations safe in dynamic, real-world environments.

NbP might enable more powerful, energy-efficient chips

Stanford University researchers discovered that niobium phosphide (NbP) outperforms copper in conductivity at thicknesses below 5 nanometres, thinner than current chip wiring. NbP's topological properties allow its surface conductivity to remain stable, even when altered or thinned, resisting performance degradation. Remarkably, NbP maintains exceptional conductivity without precise structural control, simplifying manufacturing and reducing costs. This could revolutionise chip design, offering more powerful, energy-efficient devices with reduced heat, though scalability and availability remain concerns.

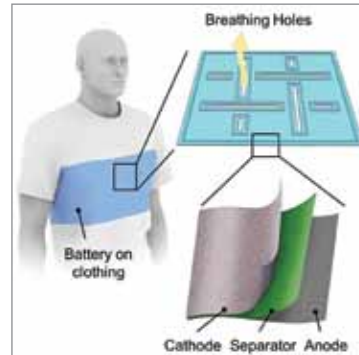
World's thinnest wave absorber made for 6G networks

In partnership with Nippon Denko, University of Tokyo researchers have created the world's thinnest electromagnetic wave absorber for 0.1-1THz frequencies, crucial for 6G deployment. The absorber, made of lambda-titanium-pentoxide coated with titanium dioxide, minimises signal interference by dissipating unwanted waves. At just 48

micrometres thick, it is durable, cost-effective, eco-friendly, and resistant to heat, water, and light. This innovation expands the terahertz spectrum for wireless communication, security, and tomographic imaging applications.

Holes in battery pouch make them more appealing for wearable electronics

A US team of engineers and materials scientists has developed a highly flexible and breathable wearable battery. By



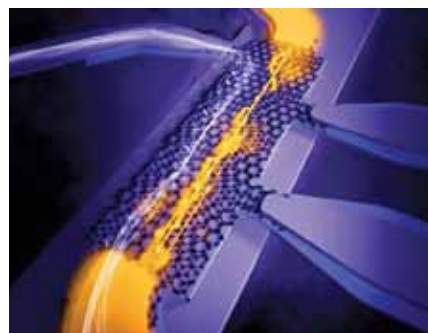
Flexible wearable batteries (Credit: <https://www.yahoo.com>)

strategically placing rectangular holes at specific locations in the battery pouch, they enhanced flexibility, stretchability (up to 10%), and breathability—making the battery twice as breathable as cotton. It withstands 100+ folding cycles without degradation, maintains charge when folded 180 degrees, and

prevents overheating, paving the way for more efficient and comfortable wearable electronics.

Breakthrough in magic-angle graphene research

MIT and Harvard researchers have measured superfluid stiffness in magic-angle graphene (MATBG), a material with



Physicists measured how readily a current of electron pairs, represented in yellow and white, flows with no resistance through 'magic-angle' graphene, represented as the black lattices (Credit: Eli Krantz, Krantz NanoArt)

unique properties due to its twisted atomic layers. This sheds light on its superconductivity, influenced by quantum geometry. Using microwave resonators, the team overcame challenges in attaching MATBG to delicate devices, measuring

shifts in resonance frequency to calculate superfluid stiffness. Their results revealed a value ten times higher than predicted, offering new insights for quantum computing technologies.